Hassan Imani

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PROFESSIONAL EXPERIENCE

Data Scientist — Financial Crime Detection — Jul 2025 – Present
Scotiabank — *Toronto, ON*

- Productionized the codebase of the primary **outlier detection model**
  - standardizing usage and accelerating development workflows across the team
  - Stabilized the training and optimization pipeline for models, achieving an average of 8% performance improvement on labeled datasets
- Re-architected the team's feature store to support scalable reuse and adaptation across projects.
- Deployed a **language model** on-prem for automating alert narrative generation

Machine Learning Engineer — Feb 2024 – Jul 2025
Zown — *Toronto, ON*

- Built the multi-stage "Matching System" multi-stage for real estate recommendations
  - Hard-constraint filtering followed by a **two-tower** model scoring user-listing affinity
  - Enriched embedding quality with auxiliary multi-task objectives predicting dwell time and engagement depth
  - Addressed cold-start via a diversified **Exploration Set** strategy.
  - Served rankings in real time: re-scored affinity per request against live user reactions
  - Automated retraining through a micro-batch pipeline gated by holdout-set validation
- Developed an **Image Tagging** pipeline supplying visual features to the retrieval model
  - Fine-tuned a **Vision Transformer (ViT)** with multi-task heads, distilling labels from a multi-modal LLM API into an owned, in-house model
- Maintained and optimized the **AI Valuation** feature
  - Trained and deployed a deep tabular model for property valuation
  - Built a **Continuous Training (CT)** pipeline automating retraining to maintain model freshness in production
- Designed a **Follow-Up Bot** and conducted a lead-conversion analysis (clustering + **PU learning**) to re-target high-potential leads

EDUCATION

York University — 2021 - 2025
PhD in Mechanical Engineering — *Toronto, ON*
- Research areas: GNNs, Transfer Learning

Tehran Polytechnic — 2018 - 2021
MSc in Mechanical Engineering — *Tehran, IR*

Sharif University of Technology — 2013 - 2018
BSc in Mechanical Engineering — *Tehran, IR*

TECHNICAL SKILLS

Languages: Python, SQL, Bash

ML / Applied: Recommender Systems, Two-Tower Retrieval, Transfer Learning, Vision Transformers, Multi-task Learning

Tools & Platforms: Git, Docker, GCP, Spark, Redis

Libraries: PyTorch, Scikit-learn, Pandas, PySpark

PUBLICATIONS

Graph Neural Networks for Sustainable Energy (HI Parashkooh, C Jian) — 2025
Predicting Adsorption in Aromatic Molecules — *ChemE 9 (4), 85*

An Innovative Method for Calculating PMV (HI Parashkooh, C Jian) — 2024
from a standard molecular trajectory — *J. Mol. Liq. 395, 123879*

Data Mining Guided Molecular Investigations (HI Parashkooh, C Jian) — 2022
on the Coalescence of Water-in-Oil Droplets — *Energy Fuels 36 (4), 1811*